

SAMYAM BHATTARAI

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EDUCATION

University of Louisiana at Monroe

May 2026

Bachelor of Science in Computer Science

Monroe, LA

- **Relevant Courses:** Data Structures and Algorithms, Database Management (SQL), Computer Architecture, Computer Networks, Object-Oriented Programming, Operating Systems, Software Development Lifecycle (SDLC), Mathematics

PROJECTS

CodeHawk — Automated Code Grading Platform with Sandboxed Execution and Plagiarism Detection

[Link](#)

Tech Stack: Next.js, React.js, TailwindCSS, Spring Boot, Java, MySQL, Python

- Architected a **3 role-based portal**, with frontend spanning **30+ pages**, with **Context API** session management enforcing **role-based route protection** across the platform.
- Integrated **12 REST API endpoints** across all **3 portals** to power core academic workflows, from enrollment to grading, **LMS export**, and deadline tracking.
- Developed a **unified grading workspace** for faculty and TAs that consolidates **automated and manual evaluation** signals into a single view, reducing UI transitions per submission.

Splitz — Multi-user Expense Tracker with Settlements and Notifications

[Link](#)

Tech Stack: Next.js, TypeScript, Convex, Clerk, Resend Email API

- Engineered a **ledger-based debt normalization algorithm** that merges offsetting balances between user pairs, eliminating redundant settlement transactions.
- Optimized database performance with **8+ indexed fields** and **full-text search indexes**, improving query response time by **20%**.
- Secured user financial data by implementing **OAuth authentication** across **7 protected routes** and enforcing **mutation-layer authorization** checks, eliminating unauthorized access.

ShopSphere — E-commerce platform with Payment Systems and Admin Panel

[Link](#)

Tech Stack: React.js, Node.js, Express, MongoDB, Mongoose, Stripe

- Architected a **RESTful API** with **10+ endpoints**, implementing **JWT authentication** and **role-based middleware** to separate customer and admin operations.
- Integrated a **dual-payment checkout** with **post-payment verification** that eliminates incomplete or mismatched orders.
- Constructed a **multi-image upload pipeline** with **cloud-hosted media**, reducing upload times by **30-40%** and enabling admins to manage a **50+ item catalog**.

Elevatepath — AI-based platform for Resume, Cover Letters, and Interview Preparation

[Link](#)

Tech Stack: Next.js, Prisma ORM, PostgreSQL, Clerk, Google Gemini AI

- Launched a **full-stack AI career platform** with **12+ pages** across resume building, cover letter generation, and interview prep, achieving a **98+ Lighthouse score** and **sub-1-second load times**.
- Embedded a **generative AI model** across **5 features**, automating career content workflows and reducing manual creation time.
- Designed a **relational schema** with **5 models** and **2 migrations**, enabling full **CRUD persistence** while preserving data integrity.

NoteGenius — AI study assistant that turns PDFs into instant notes

[Link](#)

Tech Stack: Next.js, LangChain, Convex, Google Gemini AI, Clerk

- Developed a **full-stack AI document Q&A platform** that lets end users upload PDFs and receive context-aware answers, replacing manual review.
- Engineered a **RAG pipeline** with **chunked embeddings** and **top-k retrieval**, cutting average retrieval time to **0.75 seconds** with **99% accuracy**.
- Deployed a **serverless vector database** with **metadata filtering** and **3-tier fallback search**, achieving **0.25s query latency** and **100% success rate** across test dataset.

EXPERIENCE

Student Administrative Assistant

February 2023 - August 2023

Office of Student and Professional Affairs, College of Pharmacy

Monroe, LA

- Coordinated **15 to 20 daily student interactions**, reducing scheduling delays by streamlining workflows using Microsoft Office Suite.
- Managed **20 to 30 calls and emails weekly**, improving response time by organizing requests using **Python** for file organization and email filtering.
- Processed over **40 documents monthly**, improving data accuracy and retrieval efficiency by querying and managing records using **SQL**.

ACTIVITIES & CLUB INVOLVEMENT

Association for Computing Machinery

September 2023 - May 2026

Treasurer

University of Louisiana at Monroe

- Organized **15+ members** across partner organizations to plan joint programming, growing event participation **20% year-over-year**.
- Managed budgets and venue for events serving **100+ attendees**, cutting reconciliation time by **5 days** through standardized tracking.
- Led workshops and hackathons while mentoring peers on event execution, expanding the annual event count by **15%**.

TECHNICAL SKILLS

Languages & Databases: HTML, CSS, JavaScript, TypeScript, Python, Java, MySQL, MongoDB, PostgreSQL

Frameworks & Libraries: React.js, Node.js, Next.js, Express.js, Mongoose, PyTorch

Tools & Platforms: Git, GitHub, AWS, Linux, Postman, Figma, Docker